

# SMIT ROY | JAVA BACKEND DEVELOPER

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## PROFESSIONAL SUMMARY

Java Backend Developer with hands-on experience building production-grade REST APIs using Spring Boot, Spring Data JPA, and Spring Security. Skilled in clean layered architecture, transactional workflows, and real-world integrations including payment gateways and JWT-based auth. Currently expanding into microservices architecture with Kafka, OpenFeign, and Spring Cloud for distributed systems.

## TECHNICAL SKILLS

**Languages:** Java

**Frameworks:** Spring Boot, Spring Data JPA, Spring Security, Spring Boot Actuator

**Databases:** PostgreSQL, MySQL, Redis

**Cloud & Deploy:** Docker, Render, Railway, Neon Serverless, Upstash, GitHub Actions (CI/CD)

**Dev Tools:** Maven, Git & GitHub, Postman, DBaver, ModelMapper, Swagger (OpenAPI)

**Concepts:** OOP, JWT & OAuth2, System Design, REST API Design, Concurrency Control

## OPEN SOURCE

### JWT STARTER DEPENDENCY FOR SPRING BOOT | [jwt-spring-boot-starter](#) - [GitHub \(1.0.2\)](#)

May 2026

- Built a reusable JWT authentication starter for Spring Boot with auto-configuration and zero boilerplate setup
- Implemented **JwtService**, **JwtAuthenticationFilter**, and **SecurityFilterChain** as auto-configured beans with **@ConditionalOnMissingBean** for full extensibility
- Published to **GitHub Packages** (currently with 40+ GitHub stars) with Maven and an automated CI/CD pipeline using **GitHub Actions**
- Designed a plug-and-play architecture allowing developers to override any component via custom bean definitions

## MAJOR PROJECTS

### STAYGRID | Hotel Rental Booking Backend - [Render](#) | [GitHub](#)

March 2026 - Present

- Designed and developed a scalable backend system for **property listings** and **bookings** using Spring Boot
- Implemented **JWT-based authentication** with refresh tokens and role-based access control (GUEST / HOTEL\_MANAGER)
- Developed booking logic with **date conflict resolution** and **transactional workflows**, preventing overlapping reservations across 365 days of forward inventory per room
- Integrated **pagination** and **filtering** (location, price range) to optimize API performance
- Applied **DTO pattern**, **global exception handling**, and standardized API responses for a maintainable and consistent architecture
- Integrated Redis **caching for hotel and room search results** to reduce database load on read-heavy endpoints; used **@CacheEvict** to maintain cache consistency on all hotel updates. Redis deployed on **Upstash**
- Implemented **dynamic hotel pricing** using the **Decorator Pattern** with layered surge, occupancy, and urgency-based adjustments, running on an hourly scheduler
- Integrated **Stripe payment gateway** with webhook-based confirmation and automated refund handling for cancelled bookings
- Designed a multi-state inventory management system (reserved → confirmed → cancelled) to maintain booking consistency
- Implemented concurrency control using **pessimistic locking** to prevent overbooking under multi-user concurrent scenarios
- Containerized the application with Docker and deployed it on **Render** with PostgreSQL deployed on **Neon Serverless Postgres**

### CLINIXHUB | Hospital Management System - [GitHub](#)

April 2026 - May 2026

- Built a production-grade Hospital Management REST API covering patient management, doctor onboarding, and appointment booking with a clean layered architecture
- Implemented stateless **JWT authentication** with a custom **OncePerRequestFilter** and integrated **OAuth2 social login (Google)** with automatic user provisioning
- Designed a **hybrid Role + Permission-based access control system** (ADMIN, DOCTOR, PATIENT) using Spring Security 6's **@PreAuthorize** and **@Secured** at both URL and method level
- Modelled complex JPA relationships (**@MapsId**, **@OneToMany**, **@ManyToMany**) and wrote **custom JPQL queries** for aggregation, pagination, and bulk updates
- Built a **centralized exception handling layer** using **@RestControllerAdvice**, returning consistent **ApiError** responses for auth, JWT, and access denied failures

### LMS PORTAL | Learning Management System - [Render](#) | [GitHub](#) | [YouTube](#)

January 2026 - March 2026

- Built a RESTful backend API using Spring Boot to manage **students**, **courses**, **instructors**, and **enrollments**
- Designed relational data models with JPA (**One-to-Many**, **Many-to-One** mappings) for a normalized schema
- Developed a course enrollment system with duplicate prevention logic and enrollment tracking
- Implemented input validation and global exception handling to ensure robust and predictable API behavior
- Created a **preloaded dataset (data.sql)** to simulate real-world LMS scenarios for testing and demonstration
- Containerized the application with **Docker** and deployed it on **Render** with PostgreSQL deployed on **Neon Serverless Postgres**
- Documented and demonstrated the full system end-to-end via a public YouTube walkthrough

## PROFESSIONAL EXPERIENCE

### TASK VIRTUAL SERVICES

Kolkata, India

#### Associate

March 2024 - Present

- Developed and maintained 15+ websites across clients of various domains, handling end-to-end development and delivery
- Built custom plugins and theme extensions using hooks and filters, with JavaScript handling dynamic behaviour and client-side logic across projects
- Implemented custom JavaScript for theme customizations, form validation, and front-end business logic across multiple client projects
- Integrated third-party APIs and payment gateways, managing plugin configurations, MySQL schema changes, data migrations, and query-level debugging across live projects
- Conducted end-to-end QA testing, validating cross-browser compatibility, responsive behaviour, form submissions, and payment flows
- Managed hosting configuration, domain setup, deployment pipelines, and CMS workflows, ensuring smooth go-live for each project

## EDUCATION

### CHANDIGARH UNIVERSITY

Master of Computer Applications

Specialization - **Cloud Computing**

Current Semester GPA: 9.0

Punjab, India

July 2025 - August 2027

### UNIVERSITY OF CALCUTTA

Bachelor's Degree, Commerce

Cumulative GPA: 7.0

Kolkata, India

July 2021 - August 2024